

Q. 4

1. The SSE is $\sum (Y_i - \hat{Y}_i)^2$ so

$$\sum_{i=1}^N (Y - b_0 - b_1 Z_{i1} - b_2 Z_{i2})^2$$

2. b_0 : $(Y - b_0)^2$ so $-2(Y - b_0 - b_1 Z_{i1} - b_2 Z_{i2}) = 0$

b_1 : Since it has a term that comes out
 $-2(Y - b_0 - b_1 Z_{i1} - b_2 Z_{i2})(Z_{i1}) = 0$

Same b_2 : $-2(Y - b_0 - b_1 Z_{i1} - b_2 Z_{i2})(Z_{i2}) = 0$

3. Dividing by N gives mean so

b_0 partial is SSE sum = 0
so turns into $(Y - b_0 - b_1 Z_{i1} - b_2 Z_{i2}) = 0$
SSE = 0 ✓

e.g. 2 Shown in b_1 and b_2 , sum of SSE, 2 = 0

4. Can average each value

$N\bar{Y} - Nb_0 - b_1 \sum(Z_{i1}) - b_2 \sum(Z_{i2}) = 0$
because $1/N \sum Z_{i1} = 0$, $\sum Z_{i1} = 0$
leaves $N\bar{Y} = Nb_0 \Rightarrow \bar{Y} = b_0$

so now $\sum_N (Y_i - \bar{Y} - b_1 Z_{i1} - b_2 Z_{i2}) Z_{i1} = 0$

Had to ask Gemini; here

$$5. b_1 \sum z_{i1}^2 + b_2 \sum z_{i1} z_{i2} = \sum (y_i - \bar{y}) z_{i1}$$

$$b_1 \sum z_{i1} z_{i2} + b_2 \sum z_{i2}^2 = \sum (y_i - \bar{y}) z_{i2}$$

$$\underbrace{\begin{pmatrix} \sum z_{i1}^2 & \sum z_{i1} z_{i2} \\ \sum z_{i1} z_{i2} & \sum z_{i2}^2 \end{pmatrix}}_A \underbrace{\begin{pmatrix} b_1 \\ b_2 \end{pmatrix}}_b = \underbrace{\begin{pmatrix} \sum (y_i - \bar{y}) z_{i1} \\ \sum (y_i - \bar{y}) z_{i2} \end{pmatrix}}_c$$

$$6. \frac{\sum z_{i1}^2}{N} = \frac{\sum (z_{i1} - \bar{z}_1)^2}{N} = \sigma_1^2 \text{ and } \sigma_{12}$$

$$\frac{\sum (y_i - \bar{y}) z_{i1}}{N} = \text{cov}(Y, X_1) = \sigma_{Y1}$$

$$b_1 \sigma_1^2 + b_2 \sigma_{12} = \sigma_{Y1} \quad b_1 \sigma_{12} + b_2 \sigma_2^2 = \sigma_{Y2}$$

$$\begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} \sigma_{Y1} \\ \sigma_{Y2} \end{pmatrix}$$

$$\frac{\sum (y_i - \bar{y}) z_{i1}}{N}$$

Confused here much!

$$\frac{\sum (y_i - \bar{y})(x_{i1} - m_1)}{N} = \frac{\sum z_{i1}^2}{N} = \frac{\sum (x_{i1} - m_1)^2}{N}$$

$$\frac{\sum z_{i1} z_{i2}}{N} = \frac{\sum (x_{i1} - m_1)(x_{i2} - m_2)}{N}$$